

F4SIGHT — the talk track

Word-for-word narration for the nine-beat briefing deck, plus the live terminal walkthrough. Read plain text aloud; blue lines are stage directions, not spoken.

EVENT	PROBLEM STATEMENT	TARGET LENGTH	DECK
Smart India Hackathon 2026	26187	6–8 minutes	F4SIGHT Briefing

- 01 Cold open
- 02 The failure mode
- 03 What F4SIGHT does
- 04 Architecture walkthrough
- 05 Why the evidence holds up
- 06 The hard gates
- 07 Performance numbers
- 08 Live: make demo
- 09 Close & ask

01 Cold open

~38s • deck: COVER

Every Border Out Post already has CCTV. It does two things: it shows a live picture, and it records. **Everything else is a human in a room** — and the failure mode of a tired human isn't a loud alarm. It's silence. Nobody can tell you which five minutes nobody was watching.

That's the problem we built F4SIGHT to close. Not new cameras. Not a new subscription. The cameras that are already bolted to the fence.

▸ Let the wordmark sit on screen for a beat before you speak. Don't rush the first line — it's the whole pitch in one sentence.

02 The failure mode

~40s • deck: PROBLEM

Three gaps, and none of them are a hardware problem. **Attention doesn't scale** — one operator, a wall of feeds, an eight-hour shift. **Cost doesn't scale either** — most analytics platforms charge per camera, per month, forever. And even when something is caught, **the evidence has to survive a courtroom**, not just a screenshot.

F4SIGHT is built against all three at once.

03 What F4SIGHT does

~50s - deck: SOLUTION

It bolts onto the RTSP stream that's already coming off the camera. It detects and tracks people and vehicles, scores every candidate event through a model where every number is visible — nothing is a black box — and raises an alert only after debouncing, with real evidence attached.

And just as important: three things it will **never** do. No automated response — no barrier, no siren, no dispatch. This system recommends; a human decides. No biometric enrolment — faces are matched against a curated watchlist or not matched at all. And no plaintext plate storage — if this database ever leaks, it is not a list of who drove where.

▸ This is the trust beat. Slow down on the three "does not" lines — an audience remembers restraint more than capability.

04 Architecture, one pass

~70s - deck: ARCHITECTURE

Four processes, three stateful stores, one machine. The camera's RTSP feed goes into MediaMTX, which does three things at once: it keeps recording independently — so if analytics ever crashes, the video doesn't stop — it serves the browser over HLS, since no browser can play raw RTSP, and it hands a copy to our worker.

The worker is one pipeline, eleven stages, run per frame: ingest, then an environmental quality check we call EVQM, enhancement, detection, tracking, geometry against the zone polygons, rule evaluation, risk scoring, debounce, evidence capture, and out to the sinks.

From there it fans out to Postgres with pgvector for the watchlist, MinIO for snapshots and clips, Redis for the live alert stream, and a separate anchor service that Merkle-batches evidence hashes onto the ledger. All of it surfaces through one FastAPI service — REST, WebSocket, and a verify endpoint — into the React operator console.

IF ASKED: "WHY THREADS, NOT MICROSERVICES?"

The hot path is CPU- and GPU-bound, and every heavy call — frame decode, inference — releases the GIL, so threads genuinely parallelise. One process, not one per camera, because the accelerator is a single shared resource: N processes would mean N copies of the model in VRAM.

05 Why the evidence holds up

~45s - deck: EVIDENCE

Every alert's original frame and metadata are canonicalised under RFC 8785, hashed with SHA-256, batched into a Merkle tree, and the root is anchored to a permissioned ledger. Change one byte of that evidence later, and the dashboard's own verification panel will say **tampered** — by name, not just a red X.

We didn't just build that and hope. It's tested against every Merkle tree size from one to two hundred fifty-seven leaves, and over four hundred generated signal combinations to confirm the risk score always sums exactly to its contributions.

06 The hard gates

~55s - deck: PROOF

We didn't want to claim anything we hadn't actually run. So we wrote down every external dependency this repo could not have proven for itself — nine of them — and closed as many as we honestly could, live. Six are closed with a verified run: Docker, model export, a real camera, GPU acceleration, the visual UI itself. Two are open on purpose and said so — Hyperledger Fabric needs infrastructure we're not claiming to have stood up, so it falls back to a real hash-chained ledger that already caught a real bug during our own testing. Faces are scoped out deliberately — the invariant matters more than the feature.

▸ This is your credibility slide. If the audience only remembers one thing from this deck, it should be that the

team names its own gaps before anyone else can.

07 Performance, measured

~35s • deck: PERFORMANCE

On an RTX 3050 laptop — one tier below the GPU this profile actually targets — CUDA runs the detector two-and-a-half times faster than CPU, and sustains six point four cameras at the twelve-frame-per-second target. And this exact number was reproduced live while building this demo: thirty-four milliseconds on CPU, down to fourteen and a half on GPU, same model, same machine, today.

08 Live: make demo

~60s • switch to terminal

This is the whole thing, running. One command.

```
$ git clone <repo> && cd f4sight
$ make demo
```

That's it. It installs the environment, brings up Postgres, MinIO, Redis and MediaMTX, runs migrations, seeds a demo site with real zone geometry, pulls the free model weights the first time only, and starts the worker, the API and the anchor service together. Every one of those six steps is also its own Make target — `make test`, `make bench`, `make eval` — nothing here is a black box even at the tooling level.

► Have the browser tab pre-loaded to localhost:8000 behind the terminal, so the moment `ready` prints you can alt-tab straight into a live camera tile with a detection box on it. That cut is the demo's best moment — don't let dead air sit between the command finishing and you switching tabs.

09 Close & ask

~25s • deck: CLOSE

Vigilance at the border, powered by sight. Every layer of this is free and open — no metered inference, no per-camera licence, zero marginal cost to add the next post.

